AIVOA PHARMACEUTICALS - ACTIVE PHARMACEUTICAL INGREDIENTS DIVISION

FORMAL MANUFACTURING DEVIATION & INCIDENT INTAKE REPORT

Deviation Number: DEV-2026-0412	**Date:** March 14, 2026
API Product: Atorvastatin Calcium Trihydrate API	**Batch / Lot:** 26ATC-089
Manufacturing Unit: Unit 3 Synthesis Bay	**Stage:** Stage 4 - Crystallization & Centrifugation
Criticality / Classification: Major Deviation	**Reported By:** QA Shift Investigator

1. Incident Narrative & Analytical Non-Conformance

During routine QC release testing of finished API Batch 26ATC-089, particle size distribution (PSD) analysis on the Malvern Mastersizer 3000 yielded an Out-of-Specification (OOS) result. The measured d90 value was **78.4 µm** against the registered regulatory release specification limit of **d90 ≤ 45.0 µm** (d50: 15.0 - 25.0 µm, observed: 34.2 µm). Microscopic examination revealed dense crystalline macro-agglomerates with non-uniform polyhedral habit.

2. Process Data & Distributed Control System (DCS) Historian Review

Investigation into DCS historian records (Tag: CR-301-TT-04) for Crystallizer Reactor CR-301 revealed a critical cooling rate excursion during the nucleation window. The validated batch recipe requires linear cooling from 65.0°C to 35.0°C at 0.8°C/min over 37.5 minutes. However, at 03:12 AM, chilled water modulating control valve CV-301 failed open to 85% stroke, causing an unconstrained cooling plunge at **2.4°C/min** (300% faster than validated limit). Physical inspection IR-2026-104 revealed the valve pneumatic actuator diaphragm suffered elastomeric fatigue with air leakage.

3. Maintenance & Equipment Calibration History

SAP Plant Maintenance records indicated that work order PM-ACT-08 for pneumatic actuator diaphragm replacement and loop calibration had been due on January 28, 2026. The task was deferred 45 days by the maintenance planner due to commercial production scheduling pressure. No formal Quality risk assessment or engineering deviation was initiated to evaluate the deferral.

4. Immediate Containment & Batch Quarantine

1. Batch 26ATC-089 (420 kg wet cake) was immediately tagged with physical yellow quarantine signage and locked in SAP warehouse status BLOCKED.
2. Crystallizer CR-301 was placed under Lock-Out/Tag-Out (LOTO) for pneumatic actuator rebuild.
3. Subsequent synthesis lot 26ATC-090 held at crude intermediate stage.